6. Error detection and correction using Hamming Code Concept.

Aim:

To write a program to implement error detection and correction using Hamming code concept. Make a test run to input data stream and verify error correction feature.

Error Correction at Data Link layer:-

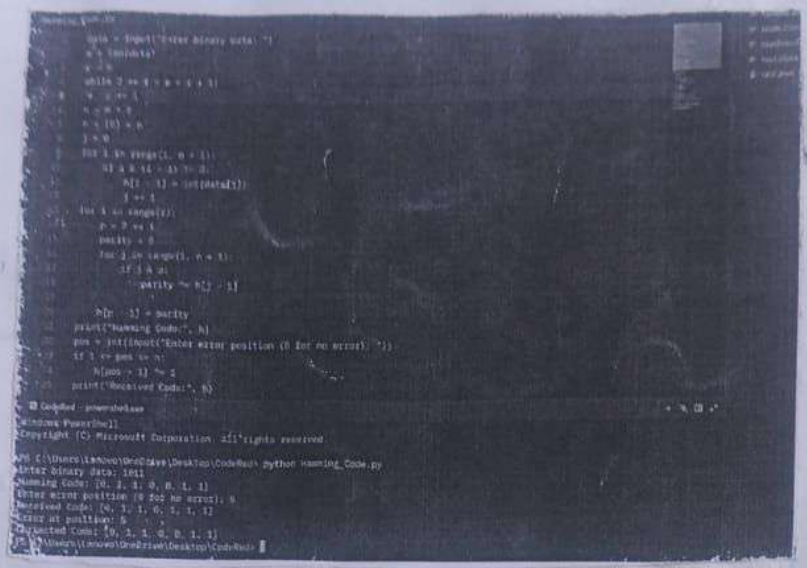
Hamming code is a set of error-correction codes that can be used to detect and correct the error that can occur when the data is transmitted from the sender to the receiver. It is a technique developed by R. W Hamming for error correction.

Create Sender program with below features:

1) Input to sender file should be a text of any length. program should convert the text to binary.

2) Apply Hamming Code Concept on the binary data and add redundant bits on it.

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```
data = input("Enter binary data: ")
n = len(data)
r = 0
while 2 ** r < n + r + 1:
    r += 1
m = n + r
h = [0] * m
j = 0
for i in range(1, m + 1):
    if i & (i - 1) != 0:
        h[i - 1] = int(data[j])
        j += 1
for i in range(r):
    p = 2 ** i
    parity = 0
    for j in range(1, m + 1):
        if j & p:
            parity ^= h[j - 1]
    h[p - 1] = parity
print("Hamming Code:", h)
pos = int(input("Enter error position (0 for no error): "))
if 1 <= pos <= m:
    h[pos - 1] ^= 1
print("Received Code:", h)
```

```
Windows PowerShell
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PS C:\Users\Lenovo\OneDrive\Desktop\CodeRun> python Hamming_Code.py
Enter binary data: 1011
Hamming Code: [0, 1, 1, 0, 0, 1, 1]
Enter error position (0 for no error): 6
Received Code: [0, 1, 1, 0, 1, 1, 1]
Error at position: 6
Corrected Code: [0, 1, 1, 0, 0, 1, 1]
PS C:\Users\Lenovo\OneDrive\Desktop\CodeRun>
```

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$P_1 = d_1 \wedge d_2 \wedge d_4$

$P_2 = d_1 \wedge d_3 \wedge d_4$

$P_4 = d_2 \wedge d_3 \wedge d_4$

code = Str(p1) + Str(p2) + Str(d1) + Str(P4) + Str(d2)
+ Str(d3) + Str(d4)

encode += code.

return encoded

text = input ("Enter text:")

binary = " ".join(format(ord(c), '08b') for c in
text)

print("Binary data": , binary)

encoded_data hamming - encode (binary)

with open("Channel.txt", "w") as file:

file.write(encoded_data)

print("Hamming encoded data :", encoded data)

print("Data Saved to Channel.txt")

Receiver program:

→ Reads data from channel.txt

→ Checks every 7 bit hamming block.

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→ Finds the error position using syndrome

→ Corrects the error or receive parity bits

→ Converts binary back to ASCII text.

Code:

```
def hamming_decode(data):
    Corrected = ""
    error_found = False
    for i in range(0, len(data), 7):
        block = list(map(int, data[i, i+7]))
        if len(block) < 7:
            Continue
    p1, p2, p3, p4, d1, p4, d2, d3, d4, block
    S1 = p1 ^ d1 ^ d2 ^ d4
    S2 = p2 ^ d1 ^ d3 ^ d4
    S4 = p4 ^ d2 ^ d3 ^ d4
error_position = S1 + (S2*2) + (S4*4).
if error_position != 0:
    error_found = True
    actual_position = i + error_position
    print("Error detected at bit position", )
```

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2) Save this output in a file called Channel

Create a receiver program with below specifics

1) Receive program should read the input from Channel file.

2) Apply Hamming Code on the binary data to check for error.

3) If there is an error, display the position of the error.

4) Else remove the redundant bits and convert the binary data to ascii and display the output.

Code:

```
def hamming-encode (data):
    encoded = " "
for i in range (0, len (data), 4),
        d = data[i:i+4]
    if len(d) < 4:
        d = d.ljust (4, '0')
d1, d2, d3, d4 = map (int, d)
```

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```
print("\n Corrected binary data :")
print (binary - data)
if error - found
    print ("\n Error was detected and corrected.").
else.
    print ("\n No error detected .")
print ( "Received text :", text ).
```

Input:

HELLO

~~Input~~ output:

Enter the text: HELLO

Error detected at position: 1

Error Corrected

Original Message: HELLO.

Result:

Therefore, the Error detection and Error Correction using Hamming code was executed Successfully.

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